Graphical Abstract

IoT-based Crop Monitoring Systems for Productivity in Smart Agriculture: a Systematic Mapping Study

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Highlights

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IoT-based Crop Monitoring Systems for Productivity in Smart Agriculture: a Systematic Mapping Study

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Abstract

Agriculture plays a vital role in ensuring global food security, requiring efficient resource management. Traditional farming often demands significant time for crop monitoring, limiting focus on irrigation and pest control. Additionally, agriculture faces major challenges including environmental degradation, declining arable land, and a projected 30% increase in food demand by 2050. Smart Agriculture (SA), supported by Internet of Things (IoT) technologies, provides promising solutions for precision resource management through IoT-based monitoring systems. This study applies a systematic mapping approach to explore IoT-based monitoring in SA and the integration of Artificial Intelligence (AI). From 342 identified publications, 108 were selected according to inclusion and exclusion criteria. Approximately 79.8% of the selected studies employed AI techniques, mainly for disease detection and smart irrigation aimed at reducing water use. Monitored variables typically included environmental factors such as humidity, temperature, and rainfall, along with soil conditions like moisture and pH. Hardware platforms were predominantly Arduino and Raspberry Pi, often deployed in fixed outdoor settings. Wi-Fi was the dominant communication protocol (42%). Deep Learning (DL), particularly Convolutional Neural Networks (CNN), was widely applied for image processing, while Machine Learning (ML) methods such as Support Vector Machine (SVM) and Random Forest (RF) were used for tabular classification tasks. An increase in Decision Support Systems (DSS) usage was also noted, highlighting their potential to enhance agricultural productivity. Overall, the study highlights the potential research branches that can further strengthen IoT and AI applications in Smart Agriculture.

Keywords: Systematic Mapping Study, Smart Agriculture, Crop monitoring, Internet of things, Decision Support Systems

1. INTRODUCTION

Agriculture is a fundamental activity to ensure global food security and plays a central role in modern discussions on agricultural sustainability. Ultimately, achieving a successful harvest requires the efficient use of natural resources and the implementation of sustainable farming practices. Achieving a successful harvest requires an understanding of the essential conditions for crop development. Traditional farmers dedicate around 70% of their time to crop monitoring, limiting the time for other critical tasks such as irrigation, fertilization, and pest control, which are essential for optimizing crop yield performance by carefully managing factors such as water, nutrients, temperature, and light [1, 2].

Agriculture has evolved considerably from traditional methods rooted in ancestral knowledge. Innovations such as mechanization, genetically modified seeds and advanced farming techniques, including greenhouse farming, vertical farming, hydroponics, and aquaponics, have significantly increased food production [3]. However, these innovations have also raised concerns, particularly with regard to the intensive use of pesticides, environmental degradation, biodiversity loss, and public health risks [4]. Smart Agriculture (SA), which integrates technologies such as the internet of things (IoT), wireless sensor networks, agricultural robots, unmanaged aerial vehicles (UAVs), and image processing, represents a transformation of agricultural practices in response to current challenges [1]. These technologies enable precise environmental monitoring, process automation, and optimal resource allocation, while artificial intelligence (AI) further enhances decision-making by leveraging data analytics to improve production efficiency, reduce post-harvest losses, and promote the sustainability of agroecosystems [5].

A key component of Smart Agriculture is the implementation of smart monitoring systems, which consist of interconnected devices and technologies capable of collecting, processing and analyzing data in real time. These systems complement the technologies integrated in SA and work together with artificial intelligence to provide critical insights for informed decision-making [6]. These systems, including sensors, communication modules, and

intelligent algorithms, enable accurate and autonomous monitoring of environmental and operational variables [7], which makes them essential in Smart Agriculture for efficient resource management. They allow for timely interventions and optimal allocation of inputs such as water, nutrients, and labor [1]. For example, these smart monitoring systems can generate automatic alerts in response to adverse conditions, optimize water usage, and adjust fertilization practices according to the specific needs of the crop. In addition, such systems provide farmers with detailed information on plant development stages, soil moisture, and local weather trends, facilitating timely interventions to maximize productivity and reduce risks [4].

Smart monitoring systems are commonly implemented on intelligent platforms capable of measuring a wide range of spatio-temporal variables. Many of these platforms integrate AI techniques such as machine learning and deep learning to predict critical outcomes, including the occurrence of diseases and pests, stages of crop development, and yield estimates. These predictive capabilities allow farmers to make proactive decisions, ultimately improving operational efficiency and minimizing potential losses [6].

This paper presents a systematic mapping study on IoT-based crop monitoring systems, with a particular focus on their contribution to enhancing agricultural productivity, including optimizing water and nutrient management [8, 9, 10], providing real-time environmental monitoring [11, 12, 13], and supporting timely decision-making [14, 15]. Using the methodology proposed by Petersen [16], this study synthesizes recent research, identifies current technological trends, and outlines future research directions to support the advancement of Smart Agriculture practices.The review covers publications from 2018 to 2024 and addresses four research questions related to IoT monitoring systems. Artificial intelligence (AI) is of particular interest because it enables advanced data analysis, supports timely decision-making, and contributes to improved productivity in agricultural systems. A total of 342 documents were initially identified, of which 108 articles were selected based on well-defined inclusion and exclusion criteria. Data were extracted in relation to the various layers of the IoT reference architecture to answer research questions, examine ongoing challenges, and draw meaningful conclusions. The results obtained following the approach highlight the transformative potential of IoT and AI technologies in agriculture and provide a roadmap for future exploration in this field.

The remainder of this paper is structured as follows: Section 2 provides background on the evolution of Smart Agriculture, the use of AI, and the integration of IoT in agricultural practices. Section 3 outlines the systematic mapping methodology, including the formulation of research questions, search strategy, selection criteria, data extraction, and analysis. Section 4 presents and discusses the findings for each research question. Finally, Section 5 and 6 identify key challenges related to productivity-focused monitoring and summarize the main conclusions of the study.

2. BACKGROUND

Agriculture has evolved from traditional small-scale practices, characterized by manual labor, organic fertilization, and crop diversity, to modern conventional systems that rely on mechanization, monoculture, and intensive synthetic inputs to maximize yields [17]. Although modern methods achieve high efficiency and meet growing food demands, they also cause soil degradation, loss of biodiversity, and water pollution [18]. In response, new technology-driven approaches are emerging: precision agriculture (PA), for instance, applies sensing, monitoring, and data-driven management to optimize inputs and reduce environmental impact; SA expands the PA approach through connectivity and automation for real-time farm control; artificial intelligence enhances decision-making with predictive analytics, early detection of pests and diseases, and optimized irrigation and fertilization schedules; and IoT architectures that integrate networks of sensors, devices, and machines, enabling continuous monitoring, data exchange, and AI-powered analytics for more efficient and sustainable farming [19].

The following subsections examine the main components of technology-driven agriculture. Precision and Smart Agriculture focus on optimized farm management through sensing, monitoring, and connectivity. Artificial Intelligence is discussed for decision-making and predictive analytics, while IoT Architectures address sensor networks and continuous data-driven management.

2.1. Precision Agriculture and Smart Agriculture

Precision agriculture (PA) seeks to address the disadvantages of modern conventional agriculture while retaining its benefits by applying technologies that optimize input use and manage crops and soil with high specificity. By

leveraging GPS, sensors, satellite imagery, and UAVs, PA collects detailed data on crop, soil, and weather conditions. Data enables the precise application of water, fertilizers, and pesticides only where and when needed, thereby increasing yields, reducing waste, and lowering costs [20]. Originating in the United States in the early 1990s to improve efficiency and minimize environmental impacts [21], PA has evolved with the emergence of IoT, cloud computing, big data, AI, and other Industry 4.0 technologies [20]. PA has given rise to Agriculture 4.0 [22], also referred to as Smart Agriculture or Data-Driven Agriculture [7].

While PA focuses on optimizing input application and crop management through instrumental technologies, SA takes a broader approach, integrating IoT, big data, AI, automation, robotics, and blockchain into all stages of agricultural production [1]. SA includes precision irrigation systems that adjust water delivery based on soil moisture, variable-rate fertilization guided by yield maps, autonomous tractors, farm management platforms that aggregate multi-source data, and blockchain solutions for food supply chain traceability [23].

Driven by IoT, SA can uncover key relationships—such as between soil moisture and yields—through sensor data analysis [4]. Empirical evidence from SA-based environments shows yield increases of 20% in wheat, 15% in corn, and 5% in soybeans, alongside 10%–20% reductions in pesticide and fertilizer use [24]. Sustainable yield assessments, measuring the ability to continuously generate economic, social, and environmental benefits, report efficiency levels of 92%–95% [24]. These results demonstrate the potential of SA to transform agricultural practices by maximizing production, enhancing sustainability, and reducing environmental impact [23].

2.2. Artificial Intelligence in Smart Agriculture

AI is transforming agriculture, with two main types of approaches: Narrow AI and General AI. Narrow AI refers to specialized software, often integrated with hardware and sensors, designed to perform specific tasks. For example, autonomous tractors in farming utilize narrow AI to operate independently. On the other hand, General AI is more advanced by resembling human intelligence with self-awareness and the ability to perform a wide range of tasks, such as complex reasoning, autonomous learning, natural language understanding, multi-sensory perception, creative problem-solving,

and advanced decision-making [25].

In agriculture, AI is increasingly applied across a wide range of tasks. Farmers collaborate with data companies to implement AI in crop monitoring for decision-making in the management of resources such as water, fertilizers, and pesticides. By collecting detailed information on plant health, pest presence, and environmental conditions, farmers can apply resources more precisely, targeting only specific areas and optimizing dosage. Notable examples of AI applications include the automation of robots that plant seeds, regulate irrigation, maintain fields, and harvest crops [1]. Overall, these automations enhance productivity while promoting sustainable practices by reducing environmental impact and improving worker safety [26]. Looking to the future, AI is expected to play a crucial role in agriculture, which remains a cornerstone of any economy [27]. While farming has traditionally focused on food production, farming has evolved over the past two decades to encompass the production, manufacturing, marketing, and distribution of agricultural products [28]. Today, agriculture is vital for livelihood, Gross Domestic Product (GDP) growth, trade, employment, and raw material supply [29].

To address the challenges of a growing global population, such as the increasing demand for food and the decreasing availability of arable land, agricultural practices need to become smarter [1]. Emerging technologies like Big Data Analytics, Robotics, IoT, affordable sensors and cameras, UAVs, and widespread Internet access in rural areas will facilitate the integration and adoption of AI in agriculture [30]. AI systems can provide predictive insights on which crops to plant [10, 15], the best times to seed and harvest [31], and how to optimize resources like water, fertilizers, and pesticides [8]. By analyzing data such as temperature, weather, soil conditions, and historical crop performance, AI can help maximize crop yields while minimizing environmental impact [6].

2.3. An IoT-Based Architecture for Smart Agriculture

An IoT architecture may vary depending on the needs of each use case. The authors in [32] provide an overview of multiple IoT architectures for agriculture and propose a more comprehensive one: a seven-layer architecture, which has demonstrated good interoperability and integration with AI

systems. The IoT architecture considered in this study is divided into the following layers:

1. **IoT devices layer:** Is composed by sensors and by any device that collects information from the environment. These devices are key for the acquisition of data that will be used in the next layers.
2. **Mobility unit layer:** This layer manages the mobility and ubiquity of IoT devices. It guarantees that devices can operate continuously and efficiently while receiving data..
3. **Network layer:** This layer is responsible for establishing connections and managing communication between IoT devices, network components, and servers. It ensures that data is reliably transmitted to the device where the next layer is executed.
4. **Scenarios layer:** Processing or analytics layer, is in charge of using IA or data analytics to extract useful information from the captured data.
5. **Service layer:** Establishes interoperability among IoT devices, analytical models, and user platforms, facilitating effective integration and communication across all system components.
6. **Extended layer:** This layer encompasses edge and fog computing, as well as cloud platforms that utilize either hub-based or on-premises processing. It enables data processing close to the source, reducing latency and enhancing overall system efficiency.
7. **Controller layer:** This layer is composed of emulation and simulation software, security protocols, and access control mechanisms. It monitors and manages the entire IoT system by leveraging these software tools and protocols.

The previously presented seven-layer architecture can be condensed into a four-layer version that encompasses the main elements while retaining a dedicated layer for data analytics (see Fig. 1). In this simplified architecture, the first layer, the Perception Layer, includes all aspects of sensing, as well as edge and fog computing with their corresponding security mechanisms. The second layer, the Network Layer, is responsible for connectivity and data storage. A dedicated Analytics Layer is maintained for data processing and analysis, while the fourth layer, the Application Layer, addresses user interfaces and application functionalities. This architecture, with its clearly defined layers, serves as the basis for formulating the research questions of the

systematic mapping study, whose methodology is presented in the following section.

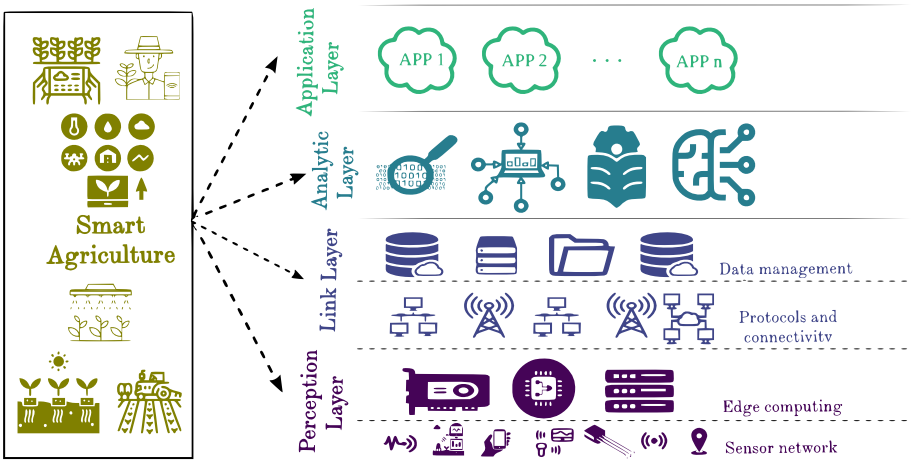


Figure 1: Four-layer IoT-base architecture for SA

3. Systematic Mapping Method

Systematic mapping is a research methodology that enables a structured and replicable collection and summarization of recent advances in a specific field. This approach provides an overview of the scientific landscape, identifying key topics, trends, and challenges [33]. Although originally developed in software engineering, its use has expanded to domains such as the Internet of Things (IoT), and in the particular case of this project, to its applications in SA.

In this paper, we describe the development of a systematic mapping study based on the methodology proposed by Petersen et al. [34]. Specific steps were defined to identify and evaluate studies on the use of Machine Learning techniques to improve productivity in IoT-based agricultural monitoring systems. A competent team was responsible for the entire methodological process, including formulating the research questions, designing and executing the search strategy, defining the selection criteria, and extracting and analyzing the data.

3.1. Research questions

We developed this study to identify the state-of-the-art on how ML techniques are used to improve the performance of IoT-based agricultural monitoring systems. Thus, we established four research questions (RQs) to guide our literature review in this field.

- **RQ1:** How is AI applied in smart agriculture platforms?
- **RQ2:** What types of AI models are utilized in smart agriculture?
- **RQ3:** How are AI models being used to improve crop productivity through an IoT system for smart agriculture?
- **RQ4:** What are the hardware platforms used to implement IoT-based smart agriculture solutions?

3.2. Search strategy

We systematically analyzed the above-stated research questions to derive the main keywords for our study: "Internet of Things", "Artificial Intelligence", "Crop", "Smart Farming", "Monitoring" and "Productivity". Next, we constructed the search queries by incorporating additional terms derived from variations of these keywords. Table 1 illustrates the search query, where each primary keyword corresponds to a defined query encompassing all its variants. The `AND` logical operator is employed to link the resultant keyword groups.

Main Keyword	**Query**
"Internet of Things"	"Internet of Things" `OR` "IoT"
"Artificial Intelligence"	"artificial intelligence" `OR` "machine learning" `OR` "deep learning"
"Crop"	crop
"Smart Farming"	"smart farm*" `OR` "precision agriculture"
"Monitoring"	monitoring
"Productivity"	productivity `OR` efficiency `OR` quality

Table 1: Search query used in the mapping study.

Based on the analysis conducted by Chen et al. [35], we selected four of the most pertinent academic databases: IEEE, Web of Science, ACM, and Elsevier, and use Scopus as a complementary meta search engine. The

search was conducted in November 2024, using the search strings shown in the Table 1 within titles, abstracts, and keywords of the published works. Subsequently, we identified a total of 342 publications.

3.3. Selection criteria

For this study, we establish five inclusion criteria and seven exclusion criteria. The inclusion criteria outlined for this study are as follows:

- IC1: Articles from the last 7 years (2018 to 2024).
- IC2: Articles sourced from the following databases: IEEE Xplore, ACM, Elsevier, Web of Science.
- IC3: Articles containing sections on Evaluation, Experiments, or Results.
- IC4: Articles that describe well-contextualized development in IoT architectures.

The criteria for excluding studies in this mapping study are as follows:

- EC1: Articles that are not well written.
- EC2: Duplicated works or previous versions of a complete study.
- EC3: Articles categorized as posters or extended abstracts, and PowerPoint files.
- EC4: Articles that do not specify a direct relationship between IoT and SA.
- EC5: Articles that do not describe solutions for SA.
- EC6: Articles with lengths less than 4 pages.
- EC7: Articles lacking evaluation, experiments, or results sections.

We apply the inclusion/exclusion criteria to evaluate the papers gathered in the preceding step, ensuring each paper undergoes analysis by all team members. We hold meetings to address any conflicts arising from the application of these criteria. We employed the Rayyan web application to manage this process effectively [36]. As a result of this process, 108 papers were finally selected.

3.4. Data extraction

During this stage, we conduct a thorough review of the papers selected to extract pertinent information to address the research questions. Following the approach outlined by Petersen et al. [16], we partition the selected articles into four sets comprising 27 articles each. Each team member extracts information from the articles within their designated set and subsequently reviews the extraction performed by another team member. Through this iterative process, we ensure that each article is reviewed by at least one team member. Subsequently, a weekly meeting is convened to resolve any conflicts and achieve a consensus.

4. RESULTS AND DISCUSSION

In this section, we analyze the data collected throughout the research, drawing conclusions to address the research questions outlined in Section 3.1. Data are presented through a series of figures and tables that visually represent the distribution and characteristics of the reviewed articles. Furthermore, we discuss the implications of these findings in the context of the research questions and identify potential limitations. Finally, the study provides recommendations for future research based on the observed gaps and trends in the literature.

The analysis of the results begins with an overview and taxonomy of the selected studies. The overview provides contextual information about the research landscape, while the taxonomy organizes the studies into dimensions and categories aligned with the research questions defined in the methodology section. Subsequently, each research question is addressed in detail to identify the main applications, types of monitoring, IoT architectures and components, as well as advancements in data analytics, artificial intelligence, and overall levels of system autonomy.

4.1. Overview and Taxonomy

From the data extraction process, a wide range of information was obtained, allowing for the contextualization and classification of the selected studies. Section 4.1 is divided into two subsections:

4.1.1. Contextual Overview

This subsection situates the reader by providing information on the publication year, country, type of study, and other relevant aspects within the mapping.

For the revision, we chose articles whose publication years fall within a seven-year period from 2018 to 2024. Figure 2 shows the distribution of the selected articles by year of publication. As expected, 2018, being the earliest year, has the lowest number of publications, with only two articles. From there, a slight upward trend is observed in the following years, reaching a peak in 2023 with a total of 45 articles. In 2024, up to the date of the search, 33 articles were retrieved, showing a moderate decrease compared to the previous year but still representing a high level of research activity. It is expected that the research trend will continue to rise by the time of publication, reflecting the growing global interest in the area of SA and the technological maturity that has enabled the reduction of IoT component costs, facilitating their adoption in agricultural environments [37]. Furthermore, global initiatives promoting sustainability and climate reflect the sustained momentum of research and investment in this field [38].

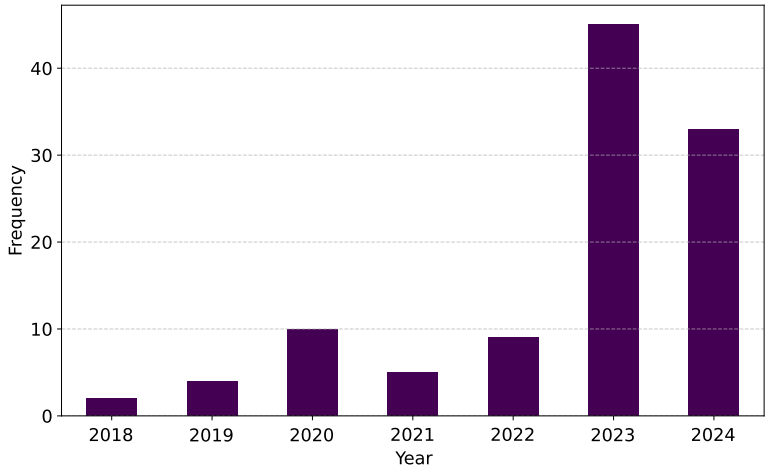


Figure 2: Publications frequency per year

The distribution of the articles according to their publication country is shown in Figure 3, where there is a clear predominance of India, which accounts for 53.7% of the publications. United States and China both rank

second, each contributing 5.6% of the total. Among the remaining countries, overall contributions are more modest, and participation levels tend to fluctuate. This distribution suggests that India has become a major reference in recent years in the field of Smart Agriculture. Figure 3 generalizes this information and highlights the countries in which at least one article has been published.

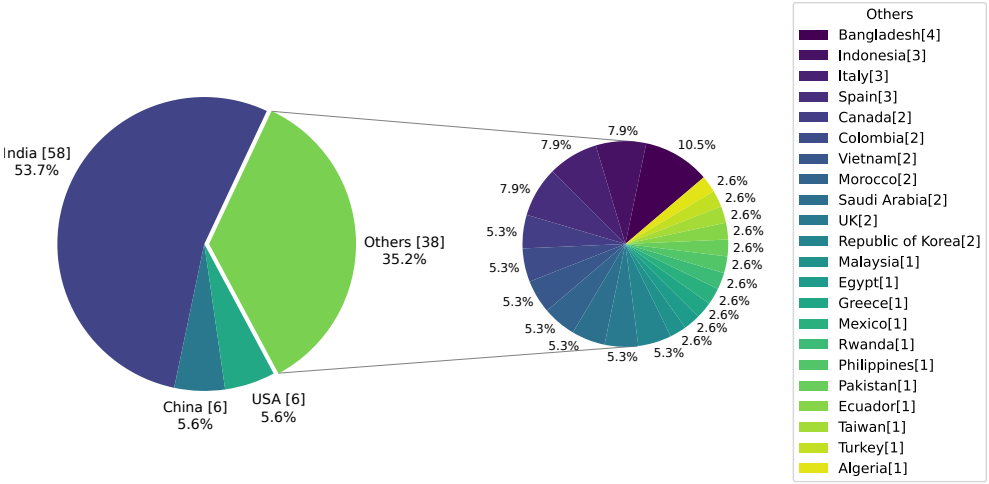
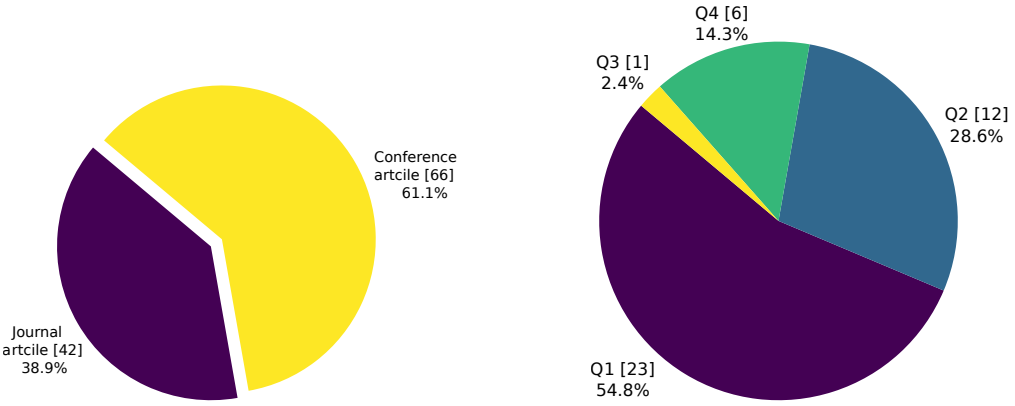


Figure 3: Documents distribution in different countries

There were two types of articles, depending on their publication type: journal articles and conference proceedings. Figure 4a shows that just over half of the articles are conference publications. In terms of impact, Figure 4b presents the distribution of article classifications based on the impact ranking of journals in their year of publication, using quartiles Q1 to Q4 as a reference, where Q1 represents the highest level in this metric. We observe that 83.4% of the journals in which the reviewed articles were published fall into Q1 or Q2 rankings, while only 16.7% have a lower rank (Q3 or Q4). This suggests that academic research in this area tends to be published in highly ranked journals, which reflects the visibility and recognition of the topic in the scientific community.

Finally, following Petersen et al. [16], research can be categorized into Validation Research, Evaluation Research, and Solution Proposals. Fig. 5 shows the classification of articles according to the type of research pre-



(a) Publication Type (b) Quartile range

Figure 4: General information

sented in them. Among the publications, 73.1% are Solution Proposals, presenting solutions that are either novel or extended without implementation. Evaluation Research represents 14.4% of the studies, investigating techniques applied in practice and assessing their benefits and limitations. Validation Research accounts for 12.5%, focusing on novel techniques that are implemented only in controlled or laboratory environments rather than in real-world or industrial settings.

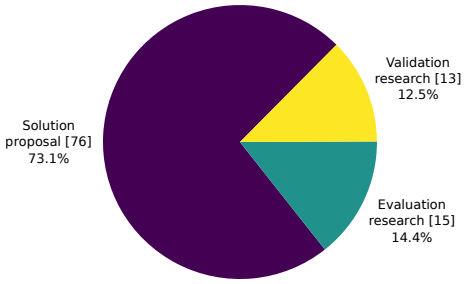


Figure 5: Research Type

Overall, the predominance of Solution Proposals suggests that most studies remain at the proposal stage, highlighting an opportunity for future research to include practical implementation.

4.1.2. Taxonomy of IoT-based Crop Monitoring Systems

This subsection presents the proposed taxonomy, defining the main dimensions of IoT-based crop monitoring systems, considering the technologies and approaches adopted, including monitoring types, use cases, IoT architecture, and AI and analytics strategies.

The taxonomy for the systematic mapping highlights the types of monitoring, as well as the use cases or applications in the agricultural domain. It also emphasizes the architecture and components of the IoT-based monitoring systems employed, as well as the data intelligence and processing approaches, indicating whether these were validated under real-world conditions or not. Table 2 presents this taxonomy divided into three main dimensions. The first dimension, Agricultural Context, includes two categories: type of research and use case. The second dimension, IoT Architecture, comprises four categories: deployment type, hardware, energy sources, and communication types. Finally, the third dimension, Artificial Intelligence and Analytics, is composed of six categories: AI implementation, model type and task, inference level (cloud, edge, others), decision support systems (DSS), and type of validation. A brief description of each category in the proposed taxonomy is presented below.

Table 2: Proposed taxonomy for the classification of studies in IoT-based smart agriculture.

Dimensions	**Categories**
1. Agricultural Reserach Context	- Use case - Type of research
2. IoT Architecture	- Processing platform - Power - Interconnection - Deployment
3. AI and Analytics	- AI implementation - Model type - Task type - Inference (Cloud / Edge) - Decision support system - Validation type

Agricultural Research Context.

- *Use Case:* Refers to the specific smart agriculture application for which different types of AI approaches have been employed.
- *Type of Research:* Research can be classified into three categories:

  Solution Proposals — Studies presenting novel or extended solutions that have not yet been implemented.

  Evaluation Research — Investigations of techniques applied in practice, assessing their benefits and limitations.

  Validation Research — Studies focusing on novel techniques implemented only in controlled or laboratory environments rather than in real-world or industrial settings.

IoT Architecture.

- *Deployment:* Describes whether end nodes are used in a fixed position, are mobile, or both, depending on the platform configuration.
- *Processing Platform:* Indicates the type of embedded system used for the IoT solution.
- *Energy Source:* Specifies the type of power supply employed to energize the end nodes.
- *Interconnection:* Refers to the communication protocol used for transmitting data collected at the perception layer.

AI and Analytics.

- *AI Implementation:* Indicates whether Artificial Intelligence is implemented in the proposed system.
- *Model Type:* Refers to the AI models and algorithms applied for decision-making.
- *Model Task Type:* Specifies whether the model's task is classification or regression.

- *Inference:* Describes the location where model inference is performed (e.g., cloud, edge, or other).
- *Decision Support System:* Indicates whether an automatic decision-support system is integrated.
- *Relationship Between Applications and Model Task Types:* Explores how model tasks relate to use cases and how this relationship impacts crop productivity.

4.2. Insights from the Research Questions

In the following sections, Research Questions (RQ1–RQ4) are systematically addressed and discussed in detail, providing a comprehensive analysis of the findings associated with each one.

4.2.1. ***RQ1:*** *How is AI applied in smart agriculture platforms?*

To determine how AI is used in SA platforms, we first identified articles that applied data analytics to process the collected data. We then analyzed what percentage of these articles specifically implemented AI models (see Figure 6). We found that approximately 79.8% of the articles implemented AI, while the rest relied on more traditional methods and techniques or did not use any at all. These results highlight a considerable increase in recent years in the use of AI for applications related to SA.

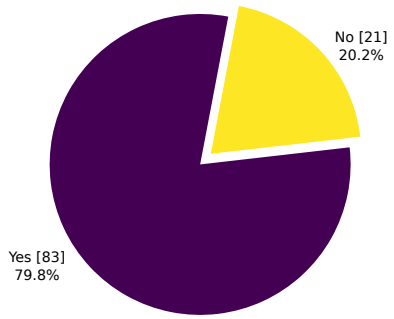


Figure 6: AI Implementation

The implementation of AI can be better understood by examining its use cases. In the reviewed articles, several applications were identified, the most common of which are disease detection in crops and the implementation of

smart irrigation systems. Figure 7 presents the identified use cases and highlights whether these were validated in some way, meaning that the authors tested the techniques employed and demonstrated that the application was effective. The use cases with the highest proportion of validations, exceeding 50% of the classified studies, are disease detection, fertilizer recommendation, monitoring, pest detection, fruit detection, and lastly, network energy use.

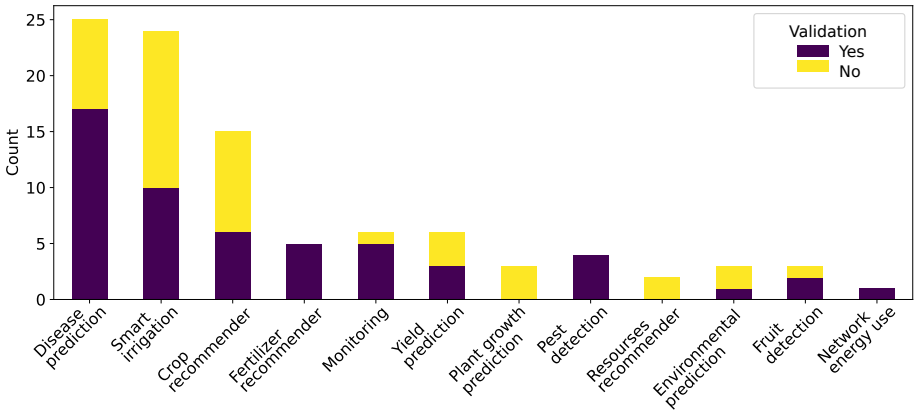


Figure 7: Purposes of implementing AI techniques in SA platforms and their validation

Depending on the specific application of artificial intelligence (AI), each study employed different groups of variables. These monitored variables represent the key factors in agriculture, providing the high-impact data required for AI models to understand, predict, and optimize crop growth and productivity. The most frequent variables; soil moisture, temperature, relative humidity, pH, and electrical conductivity, are fundamental due to their direct influence on plant physiology and environmental conditions. For instance, soil moisture regulates irrigation management, pH and electrical conductivity indicate soil health, and air temperature and humidity drive photosynthesis and disease risk [39].

The strength of AI-based analysis lies in combining these variables rather than analyzing them in isolation. Integrating soil and ambient data enables AI models to distinguish between stress factors such as drought or disease and recommend context-aware actions, such as optimal irrigation or fertilization schedules [40]. This multidimensional approach enhances model accuracy and supports more efficient and sustainable crop management [5].

Based on the multidimensional and integrated approach, the variables

extracted from the reviewed studies were systematically classified according to their measurement sources. Variables captured through soil sensors were categorized as soil data, while those obtained from environmental sensors were classified as ambient data. Data collected from irrigation-related sensors, such as water level meters, tank temperature sensors, or flow meters, are grouped under irrigation data. Variables derived from spatial sources, particularly through image acquisition techniques, are classified under image analysis. Finally, variables that did not belong to any of the previous categories but were still relevant for analysis, such as crop type, production, or cultivated area were included in the others category. As a result, five main categories were established: soil, ambient, irrigation, image-analysis-related, and others, as shown in Figure 8.

This classification is justified by the functional role that each group of variables plays within smart agriculture systems. Soil, ambient, irrigation, and image-derived data correspond to the main domains from which IoT devices collect information to characterize crop and environmental conditions [40]. Organizing variables according to their measurement sources ensures methodological consistency and facilitates the analysis of how AI models integrate multisource data to support decision-making [41]. The inclusion of the others category allows for the consideration of contextual or agronomic variables that, while not sensor-derived, provide essential information for understanding productivity and management practices. This approach aligns with previous systematic reviews and frameworks in smart agriculture [39, 42] , which also classify variables based on their origin and functional relevance within agricultural monitoring systems.

The soil category encompasses the largest number of variables and is the second most common among the reviewed studies. The most frequently monitored variables are soil moisture (70 documents, 64.8%), followed by pH (23 documents, 21.3%), soil temperature (18 documents, 16.7%), and electrical conductivity (15 documents, 13.9%). The ambient category is the most frequent and the second most diverse in terms of variable types. The most prominent variables are air temperature (83 documents, 76.9%) and relative humidity (70 documents, 64.8%), followed by precipitation (21 documents, 19.4%).

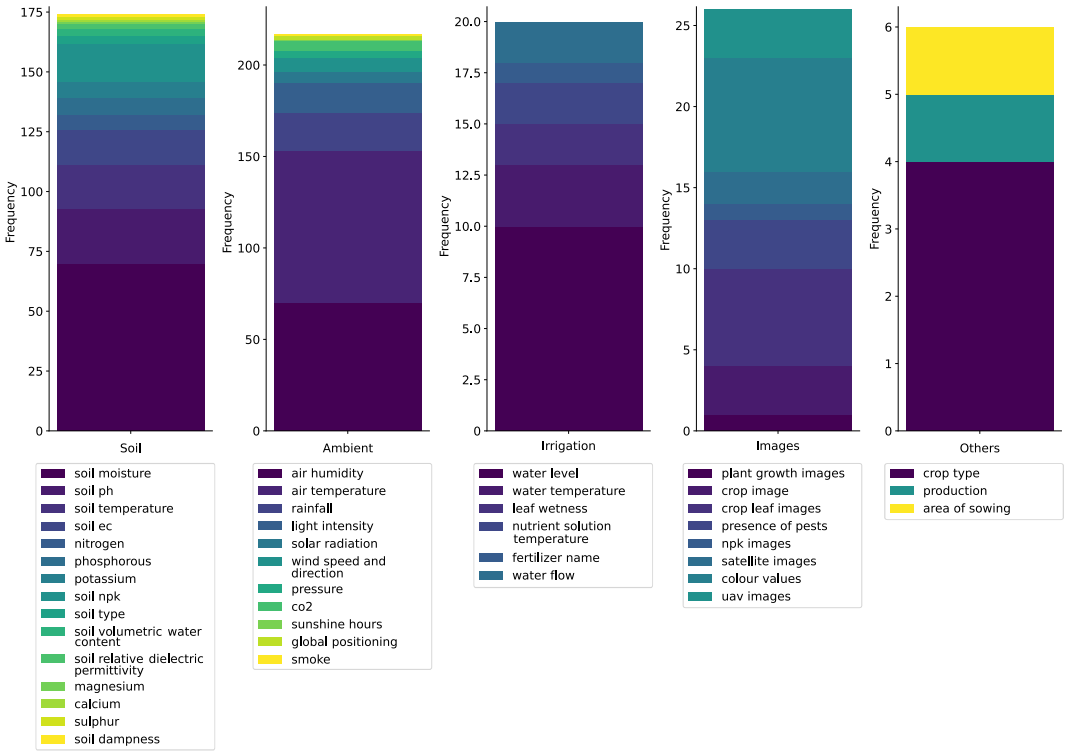


Figure 8: Categories and variables acquired in SA platforms

The "Images" category follows in quantity, highlighting the use of crop leaf images and color index values to determine crop status. The following is the "Irrigation" category, where researchers focused on water stress by measuring water levels in storage containers and monitoring the moisture content of the leaves. Lastly, it was less common to find measurements of variables such as crop type, yield, and planting area.

From Fig 8, it can be inferred that researchers who are knowledgeable about crop behavior consider understanding soil properties and climatic conditions to be crucial. These factors allow for making decisions that enhance productivity.

4.2.2. ***RQ2:*** *What types of AI models are utilized in smart agriculture?*

As highlighted in Section 4.2.1, a large percentage of the studied solutions apply AI (Figure 7), so it is necessary to obtain more detailed information about the types of algorithms used. Specifically, the AI models are

classified into Machine Learning and Deep Learning algorithms. Machine Learning models are traditionally used for processing structured data, such as relational tables. Examples of these models include Support Vector Machine (SVM), Random Forest (RF), and K-Nearest Neighbor (KNN). On the other hand, Deep Learning models are mainly employed for handling unstructured data, such as images, written documents, and more. Examples of such models include Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). The implementation of these models in the reviewed articles is presented in Figure 9. In the figure, it can be observed that Machine Learning models predominate, particularly SVM, followed by RF and KNN model. For Deep Learning models, CNN is the most commonly used, mainly for processing different types of images. ¡¡¡¡

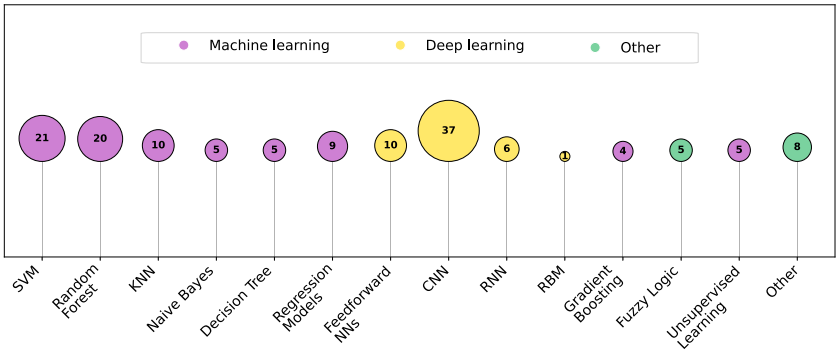


Figure 9: Models type

Figure 10 presents the frequency of classification and regression models. Classification models are used to perform diagnoses, such as assessing crop health or identifying the type of nutrient required by a plant, among others. Meanwhile, regression models are used to estimate values of various variables, such as the weight of harvested fruits based on nutrient levels, or predicting temperature over time, among others. The figure indicates that most models are classification-based, followed by regression models, while some studies utilize both types of models for their purposes.

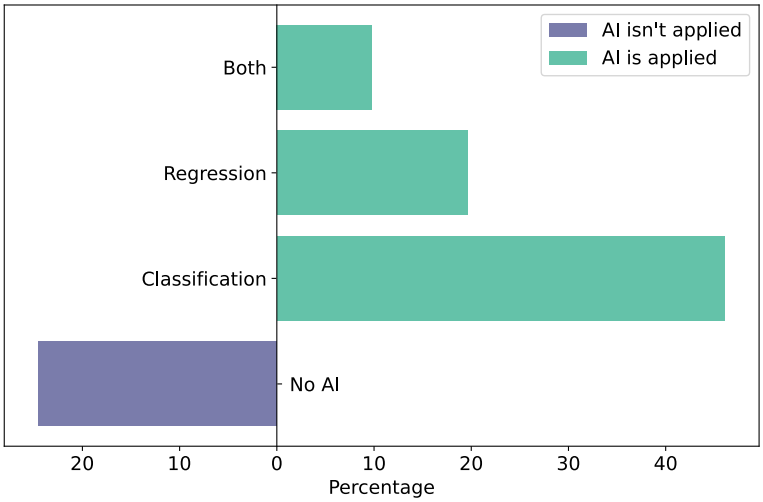
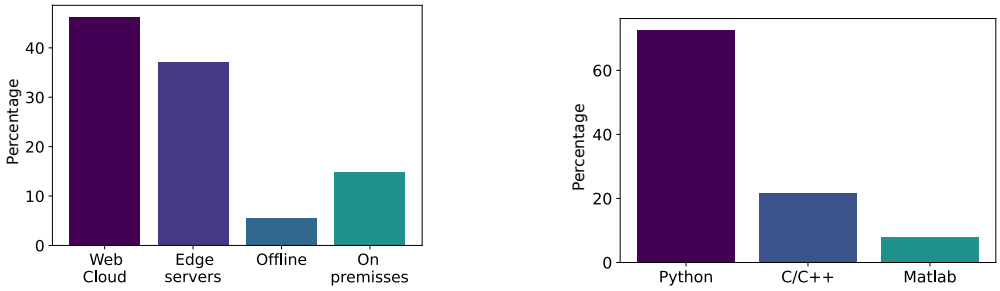


Figure 10: Model Type Applied

Finally, regarding general information related to inference environments (Figure 11a) and programming languages used (Figure 11b), it can be observed that authors prefer web cloud services for inference, followed by edge services. As for programming languages, Python remains the most widely used language for this type of solution, surpassing C/C++ and others.



(a) Inference

(b) Programming Language

Figure 11: General information

4.2.3. ***RQ3:*** *How are AI models being used to improve crop productivity through an IoT system for smart agriculture?*

Improving production in terms of yield can be approached from different perspectives. Generally, the purposes related to SA are aimed at achieving healthier, more productive, and competitive crops. Therefore, identifying

these purposes and linking them with AI algorithms helps to understand how different models can enhance the productive stage of various crops. Figure 12 illustrates the relationship between the purposes and the type of model used. It shows that disease diagnosis, yield prediction, and smart irrigation are primarily improved by researchers using classification models. However, regression models are also widely employed for smart irrigation, followed by disease prediction, fertilizer recommender and plant growth prediction.

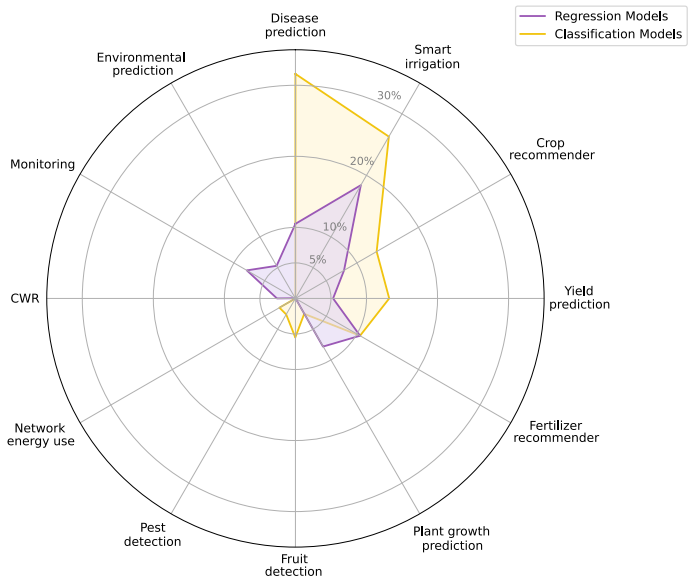


Figure 12: Purpose and AI

On the other hand, we also examined decision support systems mentioned in several studies. These systems are noteworthy as they allow farmers to receive suggestions on pesticide usage, the most suitable crop for the soil, optimal timing for irrigation and fertilization based on yield predictions, as well as providing feedback and facilitating early management planning to achieve the best quality and yield. Fortunately, the implementation of such systems has become more frequent during the last year, increasing to an adoption rate of 61.1%, as shown in Figure 13.

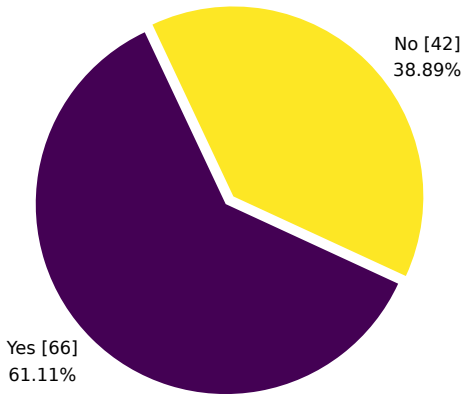


Figure 13: Decision Support systems

4.2.4. ***RQ4:*** *What are the hardware platforms used to implement IoT-based smart agriculture solutions?*

For data collection, various platforms have been utilized, as shown in Figure 14. Among the platforms used are: Arduino, Raspberry Pi, and ESP8266 microcontrollers stand out. Within the different Arduino options, low-cost platforms such as Arduino UNO have been primarily employed. In the case of Raspberry Pi platforms, the Raspberry Pi 3 has been used more frequently, while other less known hardware platforms are employed less often.

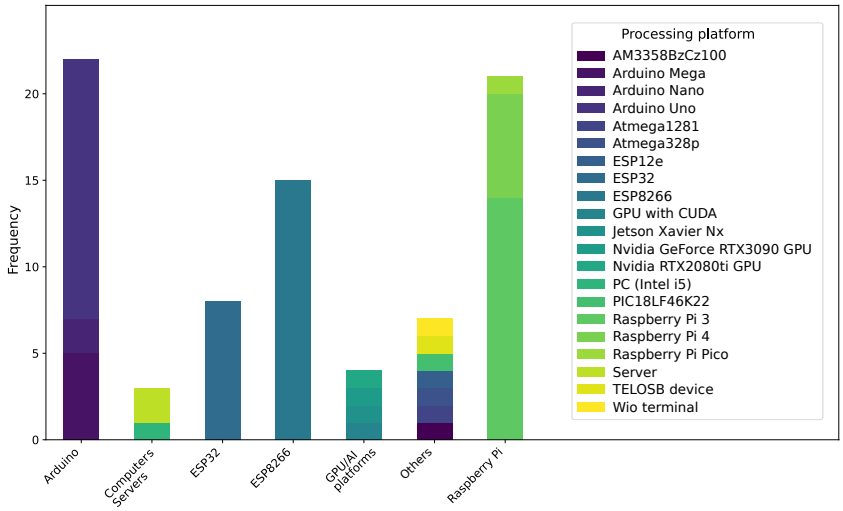


Figure 14: Processing platforms

In terms of hardware platform deployment, it was found that in 89.6% of the cases as shown in Figure 15, the deployment was carried out in a fixed manner. That is, the nodes were placed at a fixed point, capturing information exclusively at that location throughout the entire research period. Mobile platforms were deployed in 6.5% of cases, while platforms supporting both modalities (fixed and mobile) were used in 3.9%.

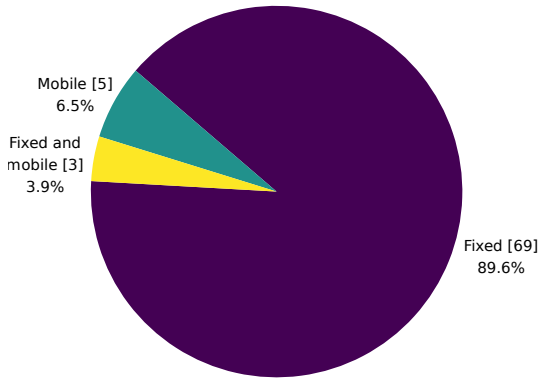


Figure 15: Deployment

To expand the information collected about the platforms, the following taxonomy describes the key aspects of the hardware platforms used in the reviewed studies. This taxonomy includes details on power sources, communication protocols, deployment types, and other relevant aspects. A total of 96 articles were identified that propose or mention the adoption of hardware platforms for measurement, of which 71 are presented in the taxonomy tables because they meet at least three of the attributes listed in the taxonomy. Another 12 articles did not meet any of these attributes [43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54]. Tables 3 and 4 present the 71 articles that describe the hardware platforms.

Taxonomy table, grouped by date, provides detailed information about the platforms used, their power sources, communication protocols, and the type of deployment employed, among others. According to the data presented, over the years of reviewed research documents, there has been a significant increase in the use of Arduino platforms up to 2023, with an average usage level of 30% in 2024. Raspberry Pi follows Arduino with a 23% usage share, showing more relevant participation in 2023 and 2024. Regarding power sources, the majority are powered by regulated wired sources (18%),

while sustainable power platforms such as solar panels are less used, with a 14% usage. In terms of interconnection systems, there is an exponential and continuous increase over time in the use of Wi-Fi, with a usage of 42%, followed by Zigbee at 10%, and LoRaWAN at 8%, although the latter two have experienced a decline in use in recent years.. Regarding deployment, fixed air platforms dominate over time (74%), with outdoor deployments accounting for 40% compared to 21% of greenhouse deployments.

Reference	Year	Processing platform					Power			Interconnection								Deployment			
		Arduino	Raspberry pi	ESP32	ESP8266	Others	Solar energy	Regulated	Batteries	WiFi	LoRaWAN	ZigBee	Ethernet	DSL	GSM/GPRS	Bluetooth	SigFox	Fixed	Mobile	Grenhouse	Outdoor
[55]	2018		✓					✓		✓								✓			✓
[56]						✓					✓										✓
[11]	2019																	✓	✓	✓	
[57]							✓								✓			✓			✓
[58]						✓					✓							✓		✓	
[59]			✓				✓									✓		✓			✓
[60]	2020					✓	✓					✓	✓	✓				✓			✓
[61]				✓					✓	✓								✓			✓
[62]		✓								✓								✓		✓	
[63]		✓									✓							✓			
[64]			✓				✓				✓							✓	✓		✓
[65]					✓					✓								✓			
[66]						✓				✓								✓			✓
[67]	2021								✓									✓	✓		
[68]			✓							✓		✓				✓	✓	✓			✓
[69]			✓				✓			✓								✓			
[70]			✓								✓							✓			
[71]	2022									✓	✓	✓				✓					
[72]																		✓	✓		✓
[73]		✓	✓					✓		✓								✓		✓	
[74]		✓					✓			✓					✓						✓
[56]		✓						✓							✓			✓			
[31]		✓							✓												
[75]					✓					✓								✓			
[76]	2023	✓					✓			✓	✓							✓			✓
[12]			✓					✓		✓								✓		✓	
[77]		✓								✓					✓			✓		✓	
[78]		✓								✓								✓			
[79]		✓								✓								✓			
[80]				✓				✓		✓						✓					
[8]		✓						✓		✓					✓			✓			✓
[10]		✓					✓	✓		✓									✓		✓
[81]										✓								✓	✓		✓
[82]				✓			✓			✓		✓						✓			
[83]					✓				✓	✓								✓			
[84]						✓			✓									✓	✓		✓
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[88]		✓							✓	✓								✓			✓
[89]			✓		✓				✓	✓								✓			✓
[90]		✓							✓									✓		✓	
[91]			✓						✓											✓	
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[93]				✓														✓		✓	
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[98]			✓						✓									✓		✓	
[99]		✓								✓		✓				✓		✓		✓	
[100]			✓							✓					✓			✓			
[15]					✓					✓								✓			✓
%		30	23	10	14	16	14	18	17	42	8	10	1	1	8	8	1	74	9	21	40

Table 3: IoT-based SA Hardware platform taxonomy (2018-2023)

Reference	Year	Processing platform					Power			Interconnection								Deployment			
		Arduino	Raspberry pi	ESP32	ESP8266	Others	Solar energy	Regulated	Batteries	WiFi	LoRaWAN	ZigBee	Ethernet	DSL	GSM/GPRS	Bluetooth	SigFox	Fixed	Mobile	Grenhouse	Outdoor
[101]	2024					✓				✓								✓		✓	
[102]			✓						✓										✓	✓	✓
[103]						✓				✓								✓			✓
[104]				✓						✓								✓			✓
[105]		✓			✓					✓								✓			
[106]		✓			✓					✓								✓			
[107]			✓				✓											✓			✓
[108]			✓				✓											✓			✓
[109]				✓				✓										✓		✓	
[26]		✓						✓										✓			✓
[110]					✓			✓										✓			✓
[111]						✓														✓	✓
[112]				✓	✓		✓	✓		✓								✓		✓	
[14]			✓					✓				✓						✓			✓
[9]				✓						✓											✓
[113]		✓	✓						✓	✓					✓	✓		✓			✓
[13]						✓												✓			✓
[114]					✓				✓	✓					✓			✓			✓
%		30	23	10	14	16	14	18	17	42	8	10	1	1	8	8	1	74	9	21	40

Table 4: IoT-based SA Hardware platform taxonomy (2024)

5. CHALLENGES

Based on the information gathered from the selected research papers, several challenges were identified regarding data collection and the implementation of IoT monitoring systems in the context of SA. Firstly, since most of the reviewed papers propose solutions, it is challenging to observe a tangible transformation in the agricultural sector towards the adoption of Industry 4.0 technologies and their benefits in terms of performance and productivity. Furthermore, access to publications that provide comprehensive information about experiments, devices, databases, methods, and algorithms, ensuring the replicability of the studies, is scarce. This limitation prevents the verification of the reported results. Lastly, few studies mention clear indicators of improvement in efficiency, quality, or productivity resulting from the implementation of the proposed methods in the investigations. These limitations, particularly those related to efficiency or productivity indicators may not only be associated with the lack of implementation but also with the significant effort required to access all relevant variables in long-cycle crops

such as coffee, avocado, or cocoa [115]. These crops take years to develop and produce commercial harvests, which occur only a few times a year, and require considerable patience to build a sufficient production history that would allow the observation of the effects of using SA.

6. CONCLUSIONS

The systematic literature mapping is a rigorous and practical method that provides an overview of IoT-based monitoring systems in SA, focusing on the studying of AI techniques, as well as machine learning and deep learning models applied to various use cases. A total of 108 documents were selected for this study, of which approximately 79.8% implemented AI techniques for different objectives.

The most common use cases were related to disease detection, a critical issue that significantly impacts crops and can drastically affect productivity, and smart irrigation systems, where significant efforts are made to reduce water consumption during crop irrigation. Additionally, the platforms employed focused on monitoring environmental variables such as relative humidity, temperature, rainfall, and solar radiation, followed by soil variables such as moisture, pH, temperature, and electrical conductivity. Some studies also emphasized image analysis, particularly for tracking plant growth and detecting diseases.

Regarding hardware platforms, the most commonly used processing devices were Arduino and Raspberry Pi. Around 74% of the studies deployed fixed systems, and approximately 40% were implemented in open-field or outdoor crops. Furthermore, 42% of the cases employed Wi-Fi for connectivity.

In terms of AI models, machine learning was utilized more frequently than deep learning, with algorithms such as Random Forest and SVM being the most popular, and classification models being the majority. Finally, decision support systems are becoming increasingly prevalent, given their importance in providing recommendations for soil use, irrigation management, pest control (including fumigation), and fertilization techniques.

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